

BOUCHET HUGO

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SUMMARY

WORK EXPERIENCE

Research Intern – Monte Carlo Simulation (Medical Physics), Beihang University, Hangzhou, China Jul 2026 - Aug 2026

- FLUKA simulation of a ^{12}C beam (260 MeV/n) on a PMMA phantom; Linux – FLUKA – Flair toolchain
- Computed dose (Bragg peak) and β^+ activity; correlated it with beam range for in-beam PET
- Benchmarked results against literature data; research conducted in English

Engineering Intern – Scientific Instrumentation, INPHYNI (CNRS), Nice Jan 2026 - Jun 2026

- Real-time sync on Teensy 4.1 (C++): 250 kHz laser, 12.5 Hz mirror, 12.5 kHz camera, jitter-free
- 250 kHz ADC acquisition via DMA; Python signal processing (FFT, Hilbert) for thin-film thickness
- Opto-mechanical redesign of a hyperspectral imaging prototype; alternating school / lab schedule

Intern – OPC Engineering Assistant, Global SA, Lyon Jul 2024 - Aug 2024

EDUCATION

2026–2029 · ENSEEIHT (Toulouse INP) – MEng, Electronics, Electrical Energy & Automation

- Core courses: digital systems, circuit theory, signal & control, electric systems, programming

2023–2026 · Centrale Méditerranée – Bachelor, Responsible Engineering & Digital Transformation

- Multidisciplinary core: physics, electronics, mechanics, computing; industrial projects
- Professional immersion every year; final year in work-study (2 weeks school / 2 weeks lab)

2022–2023 · Lycée Jean Perrin, Lyon – Preparatory classes for Grandes Écoles (PCSI)

- Intensive mathematics, physics and engineering sciences (PSI option)
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LANGUAGES

- **French** : Native language
 - **English** : B1-B2
 - **Chinese** : Beginner
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SKILLS

- Programming: Python (signal processing), embedded C++ (Teensy), Java, SQL
 - Monte Carlo simulation (FLUKA, Flair), Linux
 - Engineering mathematics and physics (mechanics, electromagnetism)
 - System modeling and control fundamentals (academic level)
 - Analytical problem-solving and structured scientific reasoning
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ADDITIONAL INFORMATION

- Driving licence
- Activities: Volleyball – High-level competitive (pre-national) , Football (Soccer) – Competitive club level, departemental to regional leagues